

WASP-52b

R-0302

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_231128 · created 2026-07-17T21:07:10+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460276.6179 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.203 +/- 0.021
Transit depth (Rp/Rs)^2	4.12 +/- 0.85 [%]
Orbital Inclination (inc)	89.93 +/- 1.51
Airmass coefficient 1 (a1)	1.19 +/- 0.036
Airmass coefficient 2 (a2)	-0.146 +/- 0.064
Scatter in the residuals of the lightcurve fit is	2.97 %
Best Comparison Star	#3 - [428, 186]
Optimal Method	PSF photometry
Transit Duration (day)	0.09 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.203 ± 0.021 vs 0.1646 (deviation 1.79σ)
Duration fit vs published	0.09 d vs 0.0758 d (deviation 5.76σ)
Mid-time O–C	-4.2 min
Depth SNR	9.4
Residual scatter	2.97 %

Light curve

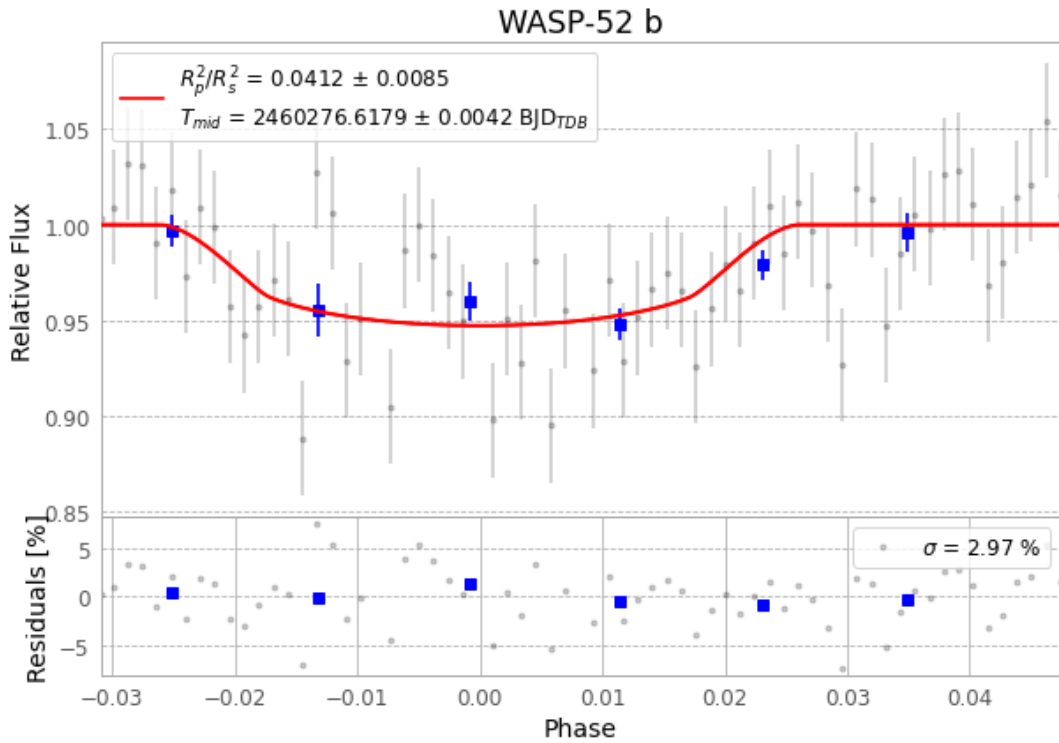


Figure 1 — FinalLightCurve_WASP-52 b_2023-11-27.png (SHA-256 3ace16268756aeb5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [318, 209], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8e1ee66527007831...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 055cf2e

Data provenance

66 FITS frames (43 MB), WASP-52231128012746.FITS → WASP-52231128044515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-231128.log (191795e11ca7...), FinalLightCurve_WASP-52 b_2023-11-27.png (3ace16268756...), FinalLightCurve_WASP-52 b_2023-11-27.csv (96540fb6bc6a...)

Compute resources

Wall time 110.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 21:07Z.